

Interventional Formation and Fracture of Spatial Referent Classes in TGI

Emylton Leunufna

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No affiliation

Abstract

An addressable crystal lattice does not itself identify which observations belong together. This manuscript defines an interventional relation operator for TGI using raw NMU action strings and before/after frames from opaque sensor ports. A one-NMU edit contrast distinguishes stable, substitution, insertion, and deletion responses without creating subword identities. Local edit transport compares changed NMU atoms across stable wrapper text while retaining the full original alignment in the proof. Ports with the same observed response history form an observational class, with C support from distinct pre-action contexts, L controls and cross-action comparators, an Omega threshold, and a unique xi lock. An actual distinguishing intervention revokes a prior class; fresh-epoch evidence is required before any successor class crystallizes. A signed three-port physical capture reproduces the sequence *NO_PATH x4, CRYSTAL, REVOKED, NO_PATH x4, CRYSTAL*. An action selector can choose an untried raw action from a finite available catalogue without pretending to know its effect. This paper states exactly what a TGI referent class means: an experience-owned relation among ports under interventions, not a target label supplied by the evaluator.

1. Research question and causal interface

The first two TGI manuscripts define static identity, coordinates, and source-valid phase. They do not decide whether two views are causally linked. The TGI goal [1] requires relation formation from experience itself. The input therefore consists of an actual action a_t and a mapping from opaque port addresses to raw frame pairs (*before, after*). The native operator sees the action characters, ports, frames, and their order. It does not receive component names, a target-port parameter, predicate labels, or the correct answer to a future query. Physical apparatus metadata may be held by the evaluator solely for independent checking.

The central contribution is a layered proof: raw edit contrast, response equivalence, support and control accumulation, phase transition, fracture, and fresh-epoch formation. Each layer retains the original event. A class cannot be asserted merely because two sensors display the same word once. The construction uses NINMENI's immutable NMU event identity [2] as a substrate, not NINMENI's trained memory tensor or optimizer [3].

2. Raw edit contrast and transport

For a port p , an event is $e_t(p) = (source_t, I(a_t), I(before_t(p)), I(after_t(p)))$, where I maps each scalar to its fixed NMU ID. The contrast operator first identifies the valid one-NMU edit kind: *STABLE*, *SUBSTITUTE*, *INSERT*, or *DELETE*. Ambiguous insertion/deletion positions are all retained. Multi-atom changes cannot be silently simplified into a one-atom response. The earlier signed three-port audit [4] is useful precisely because some inputs failed these conditions and returned *NO_PATH* rather than being selected as positive evidence.

The first response-class rule compared edit kind and absolute offset. A falsifier [5] showed it separated $T0 \rightarrow T1$ from $LONG:0 \rightarrow LONG:1$ even though both changed 0 to 1, and could equate different changed atoms at the same offset. The V4 transport rule instead uses the changed identity:

$\tau(SUBSTITUTE) = (SUBSTITUTE, old_ID, new_ID);$

$\tau(INSERT) = (INSERT, inserted_ID);$

$\tau(\text{DELETE})=(\text{DELETE}, \text{deleted_ID});$

$\tau(\text{STABLE})=(\text{STABLE}).$

The complete before/after frames, positions, and source remain in the certificate. Discarding the absolute offset from the **class key** is not discarding it from evidence. The operator makes a limited invariance claim: stable textual wrappers may change while the local NMU edit is conserved. It does not equate arbitrary paraphrases or unseen sensor transformations.

3. Class formation and phase law

Let $h_t(p)$ be the chronological sequence of certified τ values for port p . Two ports are in the same observational class at time t if their histories agree under the declared transport rule. This equivalence is operational: it says available interventions have not separated them. Members retain distinct addresses and raw histories.

For a candidate class and action, C support counts distinct raw contexts before a class-wide change. L requires (i) a same-action stable control and (ii) an appropriate comparator from another action while the candidate class remains stable. The candidate reaches an Omega support threshold only with distinct contexts. xi locks only one eligible action-class pair; ties or missing controls return *NO_PATH*. These requirements were frozen before the signed positive capture [6]. They prevent repeated identical observations or a target label from masquerading as independent support.

The phase transition is reversible. Suppose class $\{p,q\}$ is *CRYSTAL* and the next actual action changes q but not p . The event must return *REVOKED*. A V1 falsifier [7] showed that immediate promotion of $\{q\}$ from old support was wrong. The repaired law starts a new evidence epoch at the fracture. The old events remain in the certificate, but only fresh support, controls, and comparator evidence may crystallize $\{q\}$. This is a specific mechanism of source-preserving forgetting of a decision right; it is not deletion of experience.

4. Active choice without an effect oracle

After an ambiguous class crystallizes, TGI may seek a separating experience. Given a verified class certificate and a finite available-action catalogue, the V5 selector computes which action strings have already occurred in the current epoch. It preserves all untried alternatives and selects one by lexicographic NMU-ID order [8]. The catalogue is an interface affordance, not a complete ontology of world actions. The certificate explicitly reports *effect_predicted=false* and *separation_claimed=false*. If no action remains, the outcome is *NO_UNTRIED_ACTION*, not a proof that two physical entities are identical. Both source certificate and choice are independently checked.

5. Frozen physical protocol and controls

The strongest local witness is a newly created state file with initial $\{x:0,y:0,z:0\}$ and three opaque sensor ports. The collector's fixed action schedule is a,a,a,b,c,c,c,c,b . a increments x to ceiling three, b increments y to ceiling two, and c increments z to ceiling four. The sensor frames are $T(x)$, $VIEW:(x+z)$, and $D(y)$ [9]. Each action reads physical file bytes before and after; the collector records monotonic causal intervals, raw-byte hashes, a no-reset state lineage, and a signature. The evaluator's port-to-component mapping is stored apart from native input.

An independent verifier checks the Ed25519 signature, collector and executable hashes, state bytes, action effect, sensor projection, port inventory, and chronological lineage. A tampered capture is rejected. The native replay [10] produces four initial *NO_PATH* decisions, a two-port *CRYSTAL* after the same-action control and comparator, *REVOKED* on the first c , four *NO_PATH* decisions during fresh-epoch accumulation, and a one-port *CRYSTAL* after the final comparator. A wrong checkpoint digest and forged epoch are rejected; full recovery succeeds in a fresh process.

At the fifth event, a and b have already been observed; the selector chooses c from the catalogue $\{a, b, c\}$ [11]. The sixth physical event actually fractures the class. This alignment between selection and outcome is an observed consequence of this protocol, not a promise that the selector always picks a separator. The withheld edit-transport panel [12] contains 60 positive, 60 negative, and 60 wrapper/address-permutation cases across substitution, insertion, deletion, and four Unicode alphabets. Its 900 prefixes are checked by the independent oracle.

6. Results and inference boundary

The experiment establishes an intervention-owned observational relation, its revocation under actual counterevidence, and reconstruction from fresh evidence. A semantic label or designated target is not required by the operator. The physical setup still supplies the port inventory and action affordances. Two entities with indistinguishable responses to every supplied action remain one observational class until new experience separates them. This is a precise and useful referent mechanism, not a claim that the universe of possible separators was exhausted by the experiment.

The full V4 cost protocol [13] measures the changed-atom transport against the preceding offset-based class rule at equal outcome. The median ratio at 64 events is 1.012 and peak-allocation ratio is 1.032, below preregistered 1.2 limits. These are local panel costs, not end-to-end production throughput. The final source package [14] reproduces the relevant runtime and tests from extracted bytes and an isolated wheel.

7. Reproducibility and references

The primary runtime and checker are `tgi/local_edit_transport_classes.py` and `tgi/local_edit_transport_classes_check.py`; the selector and its checker are `tgi/port_distinction_exploration.py` and `tgi/port_distinction_exploration_check.py`. The signed capture, raw state, collector, verifier, and receipt are bound by hashes in the publication bundle. The sequence of failed and repaired hypotheses remains available for replication rather than being suppressed in favor of the final pass.

8. Event-level decision trace and causal reading

Event range	Raw action	Class evidence state	Native decision
1-4	a, a, a, a	distinct changing contexts and a same-action control accrue	<i>NO_PATH</i> at each prefix
5	b	cross-action comparator closes eligibility	two-port <i>CRYSTAL</i>
6	c	port responses diverge for the first time	<i>REVOKED</i>
7-10	c, c, c, c	support and a control accrue after the fracture epoch	<i>NO_PATH</i> at each prefix
11	b	comparator for the fresh candidate is observed	one-port <i>CRYSTAL</i>

The significance of event 6 is not that the letter c possesses a predefined semantic role. It is that a measured action changes one port's response history while leaving its former class mate different. If the collector had supplied $target=q$ to native input, the same status sequence would not prove endogenous class formation. The protocol instead stores the port/component mapping as evaluator-only metadata and runs a label-invariance control. Likewise, the new one-port crystal at event 11 is not inherited from the pre-fracture support: the event trace checks the epoch boundary and fresh support explicitly.

An observational class is therefore a **quotient of recorded response histories**, not a metaphysical equivalence relation over hidden physical objects. It is symmetric and transitive for the fixed history because equality of verified τ sequences is symmetric and transitive. It can refine after a new event; that change is not a contradiction in the definition because the time index has changed. Treating a past class as immutable would be

the actual contradiction. The phase lock gives the class a current reading right and the fracture law removes that right when its premises cease to hold.

9. Independent checking and attack surface

Four layers are checked separately. The Node verifier checks that the physical capture could have resulted from the recorded state transitions and that the signed body was not modified. The Python raw-contrast checker checks edit kind and all alignments. The class checker rebuilds every phase from original events, including epoch boundaries. The selector checker recomputes action coverage and deterministic choice. Agreement of all four is materially stronger than a test that compares only the final selected port with an evaluator label.

The negative controls illustrate different failure modes. The older signed panel with no same-action target control remains *NO_PATH* [4]; it cannot be rescued by post-hoc threshold adjustment. A cross-format positive that failed the offset rule motivated a changed-atom rule [5], but a different changed atom at the same offset remains negative. Forged phase or epoch fields are rejected against the raw history [10]. Duplicate available actions are rejected before selector ordering [11]. A catalogue containing only already tried actions returns *NO_UNTRIED_ACTION*; adding an untried action changes the available experiment, not past evidence. These controls bind the precise scope of the positive physical result.

References

- [1] *KONSEP.txt* and *contracts/M1_INTERVENTION_CORRESPONDENCE_RESEARCH_V1.md*.
- [2] E. Leunufna, *NMU: One Character, One Identity*, revision V2, 2026.
- [3] E. Leunufna, *Crystal Memory for Native Character-Identity Streams*, revision V3, 2026.
- [4] *evidence/m1_signed_three_port_v2_audit/terminal_receipt.json*.
- [5] *evidence/m1_cross_format_edit_v3/terminal_receipt.json*.
- [6] *contracts/M1_SIGNED_THREE_PORT_INTERVENTION_PROTOCOL_V1.md*.
- [7] *evidence/m1_unlabeled_port_fracture_v1/terminal_receipt.json*.
- [8] *contracts/M1_PORT_DISTINCTION_EXPLORATION_V1.md*.
- [9] *contracts/M1_SIGNED_PHYSICAL_FRACTURE_PROTOCOL_V1.md*.
- [10] *evidence/m1_signed_physical_fracture_v1/terminal_receipt.json*.
- [11] *evidence/m1_port_distinction_exploration_v1/terminal_receipt.json*.
- [12] *evidence/m1_v4_heldout_transport/terminal_receipt.json*.
- [13] *evidence/m1_v4_transport_cost/terminal_receipt.json*.
- [14] *evidence/source_distribution_20261004T084916022662Z/result.json*.